Alqama Shaikh

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Education

Veermata Jijabai Technological Institute 2021 – 2025
Bachelor of Technology in Mechanical Engineering – CGPA: 8 *Mumbai, India*
Coursework: Computational Fluid Dynamics, Heat Transfer, FEM, RPD, Design of Mechanical Systems, Thermodynamics (Grade: 10/10), Material Science (Grade: 9/10), Fluid Mechanics, Manufacturing Processes (Grade: 9/10), Engineering Mathematics, Applied Physics (Grade: 10/10), CAD Modelling, Robotics, Control Systems, Computer Programming (Grade: 9/10)

Extracurricular:

- Led a team to create a communication bot in 3 days for rescuing 40 workers during a construction hazard with L&T Corporation.
- Co-led a 20+ member team in knowledge transfer for robotics, embedded systems, and machine vision via workshops, seminars, and mentorship under the Eklavya program. Mentored first-year students on Dynamixel MX64 motor control, quadruped leg design, and gait analysis.
- Organized and taught DH parameters, hardware assembly, and ROS-based workshops for 50+ students in MARIO.
- Instructed IMU operation and sensor fusion to 200+ students in Wall-E. Designed convolution and filtering content for SRA's image processing seminar, Pixel.

Entrepreneurial Experience

Eyecandy Robotics June 2025 – Present
Co-founder (Hardware) – Launch Video – Blog *India*

- Built Mr. Tensaurs, Mini Tensaur, a replica of Disney's BD-X droids, and our own original character Piko – all within 8 months of starting up.
- Grew the team to 20+ people within one year and raised USD 500K+ in pre-seed funding from Founders, Inc. and Lightspeed India.
- Managed the hardware team and unblocked it across actuators and character-to-CAD design pipelines (including building an in-house CAD tool for these pipelines); set up electronics development workflows, led hiring, and delivered fun characters fast.

Research Experience

Electric Machines and Electromagnetics Research Group June 2024 – August 2024
Research Intern – Advisor: Dr. Himavarsha Dhulipati *Ottawa, Canada*

- Conducted a comprehensive analysis of existing adaptive control techniques for soft robotics.
- Designed an MPC controller from scratch for non-linear systems.
- Compared the adaptive control algorithm with a traditional PI controller in MATLAB Simulink.

Adaptive Computing Laboratory, NUS School of Computing July 2023 – October 2023
Research Intern – Advisors: Dr. Kotaro Hara, Dr. Zhao Shengdong, and Dr. David Hsu *Singapore*

- Part of the team that designed and implemented a collaborative and assistive modality for visually impaired people to navigate multi-terrain and dynamically changing surroundings.
- Devised trajectory interpretations using LiDAR-based 3D SLAM (Simultaneous Localization and Mapping).
- Conducted quantitative analysis to compare existing navigation methods with our approach.
- Conducted user surveys and experiments for iterative design.
- Continuously refined hardware components based on user feedback and data acquisition challenges.
- Co-authored and submitted a research paper to CHI 2024 as part of the collaborative project.
- Implemented ORB-SLAM3 for the Tello drone with a ROS interface and contributed to the Universidad de Zaragoza ORB-SLAM3 repository.

Competitions

Mass Robotics' Form and Function Challenge 2024 Feb 2024 – May 2024
Team Leader – GitHub Link

- Implemented 2 DOF per limb with integrated thrusters for drone functionality, enabling versatile maneuverability. Manufactured the bot using laser-cut and 3D-printed parts, and utilized Pixhawk and SpeedyBee V405 v3 for flight control.

e-Yantra Robotics Competition (eYRC), IIT Bombay – Semi-finalists Aug 2022 – Mar 2023
Team Leader – GitHub Link

- Constructed a "HolA Bot" (Holonomic and Art Bot) for use in an arena that represents an abstraction of a multimodal smart city.
- Made a three-wheeled holonomic drive controller in Embedded C using FreeRTOS.
- Used an OV5647 fisheye camera to localize with the help of ArUco markers.

Projects*

Acti-V-Link: An Active Surface Gripper for In-Hand Manipulation Apr 2023 – Mar 2024
Project Website – Tech Stack: Control Systems, Robotics, and Mechanical Design

- Designed a cost-effective gripper using SolidWorks and 3D printing with PLA material.
- Integrated an ESP32 microcontroller and FreeRTOS for precise motor control.
- Incorporated ArUco markers for visual feedback during object handling.
- Conducted experiments validating the gripper's ability to handle various objects with accurate rotational and translational movements.
- Optimized gear and linkage mechanisms for efficient manipulation.
- Maintained mechanical stress within safe limits using torque feedback.

Anuvadak – A Collaboration with L&T Construction Ltd. Sept 2023
Tech Stack: Control Systems, Robotics, Mechanical Design

- Designed a pipe-traversing bot with 6 actuators and a Raspberry Pi 3B to control PWM signals through a TB66 motor driver. Fully developed in 3 days across two versions of the bot.

Control System and Simulation of a Hexacopter with Tilt Rotors Aug 2022 – Sept 2022
GitHub Link – Tech Stack: Control Systems, Robotics, Simulation, ROS, Gazebo, Python

- Developed and implemented an optimal PID control system for an overactuated hexacopter with co-axial tilt-rotors using Python and ROS (Gazebo).
- Simulated successful omnidirectional flight.
- Reviewed relevant literature to derive allocation matrices for control.
- Performed mathematical operations from scratch in Python, including the Moore–Penrose pseudoinverse and solving linear equations.

Technical Skills

Programming Languages: Embedded C, C++, C#, Python, JavaScript, HTML, CSS
Frameworks: ROS 1 & 2 (Robot Operating System), Gazebo, RViz, OpenCV, FreeRTOS for ESP32
Software: Fusion 360, Onshape, SolidWorks, Ansys, Fluent, KiCad, VS Code, Unity Game Engine, MATLAB
Technologies: Git/GitHub, Markdown, LaTeX, Office, Linux, Bootstrap, MongoDB, Vercel, CMake, GNU Make, uv, MuJoCo, Isaac Sim
Hands On: 3D Printing, Laser Cutting, Lathe, Milling, Shaping, Rapid Prototyping, CNC Programming, Schematic Design, Circuit Bring-up and Deployment
Other: Adobe Creative Cloud (Photoshop, Illustrator, InDesign, XD, Premiere Pro)

Publications

- Shaojun Cai, Ashwin Ram, Zhengtai Gou, **Mohd Alqama Wasim Shaikh***, Yu-An Chen, Yingjia Wan, Kotaro Hara, Shengdong Zhao, and David Hsu. Navigating real-world challenges: A quadruped robot guiding system for visually impaired people in diverse environments. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI '24, New York, NY, USA, 2024. Association for Computing Machinery.
- Karthik Swaminathan, Saad Hashmi, **Alqama Shaikh***, and Vikas Phalle. Next-gen manipulation: An active surface-based underactuated gripper with visual feedback. In *2nd Workshop on Dexterous Manipulation: Design, Perception and Control (RSS)*, 2024.
- Karthik Swaminathan, Mohammed Saad Hashmi, **Alqama Shaikh***, and Vikas Phalle. Acti-V-Link: An active surface, visual feedback based, mechanically underactuated gripper for in-hand manipulation. In *Proceedings of the 2024 4th International Conference on Robotics and Control Engineering*, RobCE '24, pages 14–18, Edinburgh, UK, 2024. Association for Computing Machinery.

*Additional projects: MARIO (Manipulator on ROS-based Input/Output) on ROS2 and Implementing Path Planning using BFS on TurtleBot3.